

SpeakerScope

Live speaker-attributed transcription - native C++ desktop app
AssemblyAI × lablab.ai Hackathon - September 2026

The Problem

Meeting transcription answers *what* was said but not *who said it*. Cloud diarization services exist, but they are slow (minutes after upload), cost per minute, and send raw audio off the device. Real-time use cases - live meeting notes, call-center assistance, accessibility - need speaker labels **as people speak**, on commodity hardware.

The Solution

SpeakerScope is a Windows desktop application (C++20) that fuses two best-in-class engines into one live, color-coded transcript:

- **AssemblyAI Universal streaming** - real-time speech-to-text over WebSocket (16 kHz PCM, word-level timestamps, sub-second partials). This is *what was said*.
- **NVIDIA Nemotron-3-Diarization on-device** - the 100M-param model running locally in native C++ via the audio.cpp GGUF runtime (q8_0, no GPU, no Python). This is *who is speaking when*.

A custom alignment engine fuses both streams in the timestamp domain: every AssemblyAI word is assigned to the speaker channel active at its midpoint, producing a live, speaker-colored transcript with **Person 1...Person 8** lanes the user can rename, filter, and export.

Architecture

AUDIO	mic / WASAPI loopback / WAV -> 16 kHz mono ring buffer (SPSC, lock-free)
STT	AssemblyAI v3 WebSocket -> 50 ms pcm_s16le frames -> partial+final turns, word timestamps
DIAR	audio.cpp Nemotron-3 -> streaming inference -> open speaker turns, 8 arrival-order channels
FUSE	word-midpoint argmax + hysteresis + overlap flag -> attributed segments
UI	Dear ImGui -> live speaker lanes, editable names, per-person filter/copy/export

Engineering highlights

- **Live open-turn streaming** - patched the audio.cpp decoder to emit open (still-growing) turns mid-stream, so speakers are labeled while they talk instead of only after they pause.
- **Tuned streaming profile** - custom chunk geometry (chunk_len=40) runs **7.6× realtime** on a 12-core CPU vs. 0.53× for the default 'low' profile, with identical accuracy - no backlog, ~5 s label granularity.
- **Tee architecture** - STT and diarization consume separate rings; slow local inference can never starve the realtime AssemblyAI stream.
- **Beyond 8 speakers** - Nemotron's 8-channel limit is a concurrency limit, not a total. A GlobalSpeakerResolver layer (voice embeddings + online centroid clustering) maps recycled channels to unlimited global Person identities.
- **Privacy story** - diarization is fully on-device; only audio needed for transcription goes to AssemblyAI.

Verified results

- 117/117 automated tests green - incl. an 80-case generated-meeting battery over the turn timeline + attributor.
- End-to-end simulation: 55.8 s multi-voice wav -> live Nemotron turns + real AssemblyAI streaming -> correctly attributed, overlap-flagged transcript (speakerscope_sim.exe).
- Headless benchmark tool (speakerscope_diar_bench.exe) measures model throughput without burning API calls.

Business value

Meeting notes that say *who* said *what* are the difference between a transcript and a record. Target users: hybrid teams (meeting recaps), call centers (agent/customer separation for QA), accessibility tools, and interview/research workflows. Because diarization runs on-device, the product works in privacy-sensitive settings where raw audio cannot leave the machine - and AssemblyAI's streaming STT keeps the words best-in-class.

Stack

C++20 · CMake/vcpkg · Dear ImGui + GLFW · miniaudio · IXWebSocket + mbedTLS · AssemblyAI Streaming v3 · audio.cpp (GGUF) · Nemotron-3-Diarization q8_0 · GoogleTest · Windows 10/11, MIT license